

Soft Spaces Phase 3.2

Closing the 12-Qubit Robustness Test

Degeneracy-closed virtual-path interference and hotspot organization

Final technical note for v32.16.1

Purpose

This note closes the 12-qubit extension of the Phase 3.2 cross-qubit robustness study. The test asks whether the local hotspot mechanism identified at lower system sizes survives at 12 qubits when the analysis is formulated in a basis-invariant, degeneracy-closed way.

The central question is whether hotspot prominence remains associated with coherent interference among virtual Q -space pathways, rather than being an artifact of a particular qubit count, random Hamiltonian realization, or basis choice inside degenerate subspaces.

Frozen model and local descriptors

For an energy eigenspace projector P and complementary projector $Q = I - P$, the effective second-order feedback is decomposed over complete degeneracy-closed Q -eigenspaces P_G :

$$\Sigma_E = \sum_G T_G, \quad T_G = g_\eta(E - E_G) PVP_GVP. \quad (1)$$

The grouped interference descriptor is

$$A_E^{(G)} = \frac{2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K)}{(\sum_G \|T_G\|_F)^2} \quad (2)$$

and the grouped coherence factor is

$$C_E^{(G)} = \frac{\|\sum_G T_G\|_F}{\sum_G \|T_G\|_F}. \quad (3)$$

Positive $A_E^{(G)}$ indicates net constructive cross-channel interference; $C_E^{(G)}$ measures how coherently the grouped virtual contributions add. Both are basis-invariant under rotations inside each complete degenerate Q -eigenspace.

The 12Q run used the same frozen 11-term Hamiltonian setting and the same seed series 25042000–25042011 as the previous robustness studies. Two perturbation families were retained: dephasing (Z/ZZ) and transverse (X/XX). Spearman's ρ was used to rank the association between each local descriptor and hotspot prominence.

12Q seed-by-seed results

Seed	Groups	Dephasing ρ			Transverse ρ		
		A_G	C_G	cross	A_G	C_G	cross
25042000	32	0.5735	0.3118	0.3706	0.4324	0.4500	0.4441
25042001	64	-0.0858	0.1470	-0.0403	0.3317	0.4293	0.1103
25042002	16	0.4762	0.4524	0.6190	0.5714	0.5952	0.5238
25042003	64	-0.0235	-0.0814	-0.1056	0.1899	0.2214	0.1254
25042004	32	0.5529	0.6794	0.5794	0.4765	0.4206	0.4824
25042005	32	0.4618	0.7235	0.3324	0.3441	0.4118	0.2941
25042006	64	0.2034	0.2991	0.1745	0.5880	0.7163	0.3673
25042007	16	0.4762	0.6190	0.4286	0.0714	0.2381	0.2143
25042008	64	-0.0389	0.1888	-0.0374	0.5927	0.7801	0.5158
25042009	32	0.0471	0.2441	0.0353	0.7294	0.8147	0.5647
25042010	64	0.5066	0.7430	0.6595	0.0059	-0.1096	-0.0099
25042011	64	0.6279	0.7049	0.5110	0.3255	-0.1529	0.2441

Aggregate 12Q statistics

Descriptor	Median ρ	Positive seeds	One-sided sign-test p
Dephasing A_G	+0.4690	9/12	0.0730
Dephasing C_G	+0.3821	11/12	0.00317
Dephasing cross-sum	+0.3515	9/12	0.0730
Transverse A_G	+0.3883	12/12	0.000244
Transverse C_G	+0.4250	10/12	0.01929
Transverse cross-sum	+0.3307	11/12	0.00317

The strongest directional result is transverse A_G : all 12 seeds are positive, with median $\rho = +0.3883$. Dephasing C_G is also highly consistent, with 11/12 positive seeds and median $\rho = +0.3821$. The dephasing A_G sign test is weaker (9/12, $p \approx 0.073$), so it should not be presented as independently decisive at 12Q.

Cross-qubit context: 7Q–12Q

The 12Q results extend the previously observed finite-size pattern rather than breaking it.

Qubits	Deph. A_G	Deph. C_G	Trans. A_G	Trans. C_G
7Q	+0.4571 (11/12)	+0.4353 (11/12)	+0.2519 (9/12)	+0.4307 (9/12)
8Q	+0.2605 (10/12)	+0.2500 (12/12)	+0.2843 (10/12)	+0.5194 (12/12)
9Q	+0.3859 (10/12)	+0.4412 (11/12)	+0.2505 (11/12)	+0.3179 (11/12)
10Q	+0.3576 (10/12)	+0.5138 (12/12)	+0.4044 (11/12)	+0.4361 (11/12)
11Q	+0.2466 (9/12)	+0.3469 (10/12)	+0.2905 (9/12)	+0.3647 (11/12)
12Q	+0.4690 (9/12)	+0.3821 (11/12)	+0.3883 (12/12)	+0.4250 (10/12)

No monotonic deterioration with increasing qubit number is visible across the tested range. The effect varies from seed to seed, as expected, but the positive direction persists across system sizes and both perturbation families.

Interpretation

Phase 3.2 now supports a two-layer picture:

$$\boxed{\text{global symmetry backbone} + \text{local virtual-path interference/coherence} \longrightarrow \text{hotspot organization}} \quad (4)$$

The global layer is algebraic: the previously identified chiral-like operator Γ organizes exact $\pm E$ pairing and the corresponding transformation of the effective feedback. The local layer explains why symmetry-paired sectors do not all become equally strong hotspots. Their prominence is modulated by the way degeneracy-closed virtual pathways add coherently or destructively in Q space.

The present result is therefore stronger than a single-model correlation but weaker than an asymptotic theorem. It establishes finite-size robustness over 7–12 qubits under the frozen model family. It does not prove that the same effect must persist for arbitrarily large systems, nor does it provide a deterministic closed-form amplitude law for every hotspot.

Frozen conclusion

Phase 3.2 conclusion for v32.16.1:

Across system sizes from 7 to 12 qubits, hotspot prominence remains positively associated with basis-invariant, degeneracy-closed interference and coherence among virtual Q -space pathways. The association survives changes in Hilbert-space dimension, degeneracy structure, Hamiltonian realization, perturbation family, and random seed. Within the investigated finite-size range, no systematic weakening with increasing qubit number is observed.

This closes the 12Q robustness test. Further progress should focus on analytic reduction and documentation rather than simply extending the same scan to a larger qubit count.